

# Predictive Maintenance with Explainable AI

Interactive Predictive Maintenance Dashboard powered by  
Advanced Machine Learning & XAI



## HSA

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# PROJECT OVERVIEW & DATASET

## ⚠️ Problem Statement

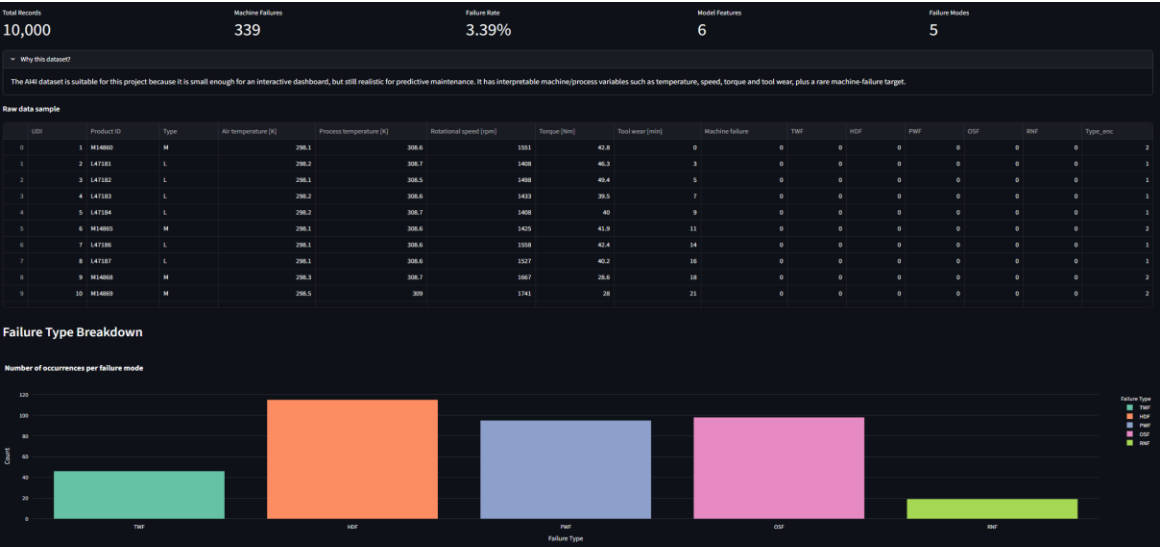
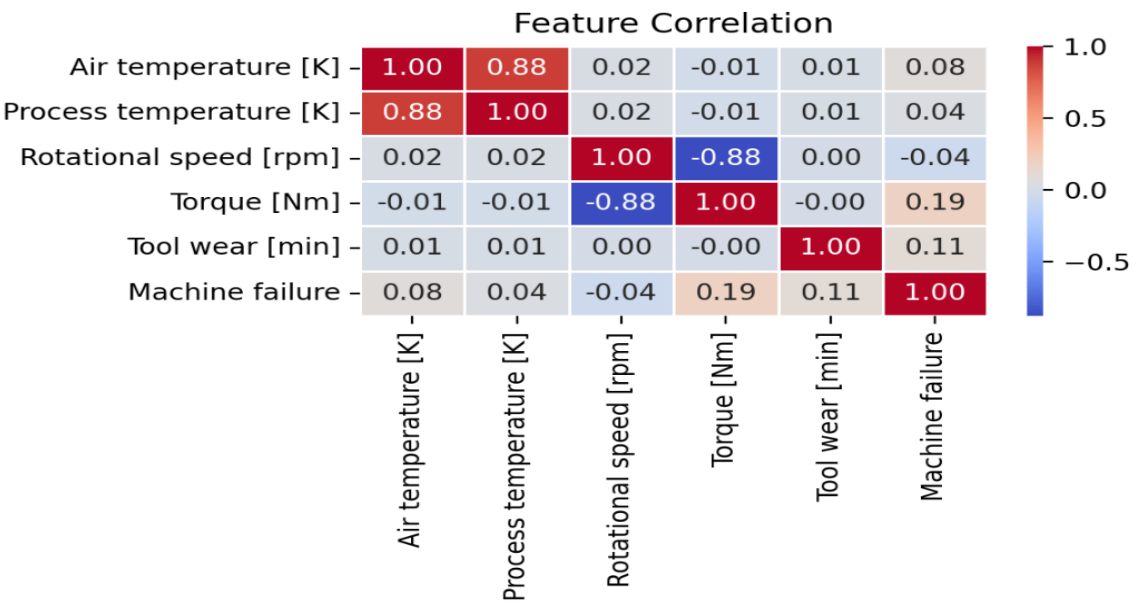
Industrial machines fail unexpectedly, resulting in massive **financial losses** and costly, unplanned operational downtime.

**Goal:** Build an interactive pipeline to predict failure risk from live sensor telemetry and diagnose root causes immediately.

**Solution:** Supervised baseline & anomaly detection models trained on raw telemetry.

## 🗄️ AI4I 2020 Dataset Overview

- ✔️ **Structure:** 10,000 operational rows, 14 continuous/categorical features
- ✔️ **Sensor Modalities:** Air/Process Temp, Rotational Speed, Torque, Tool Wear
- ✔️ **Imbalance Challenge:** Anomaly failure rate is only ~3.4% (Highly imbalanced)
- ❌ **Failure Modes:** TWF, HDF, PWF, OSF, RNF (Withheld as inputs)



# ACADEMIC DEFENSE — XAI METHOD JUSTIFICATION

How the literature supports our supervised explanations and anomaly-specific local explanations

## Defense Logic

- 1

**Supervised models**  
Output: failure probability  
XAI: SHAP • LIME • PDP
- 2

**Anomaly models**  
Output: anomaly score  
XAI: deviation / reconstruction error
- 3

**Anti-leakage rule**  
TWF/HDF/PWF/OSF/RNF displayed only,  
not model inputs

**Takeaway: global trust + local diagnosis  
+ anomaly-specific reasoning.**

## Literature → Implementation Mapping

| Project choice                           | Paper support                                                                                | Defense in our dashboard                                                                         |
|------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| AI4I dataset +<br>feature-deviation idea | Matzka (2020)<br>AI4I predictive-maintenance dataset; feature-based explanations             | Justifies our dataset, physical features, rare failures, and anomaly local explanations.         |
| LIME local surrogate                     | Torcianti & Matzka (2021)<br>LIME applied to AI4I predictive maintenance                     | Supports selected-index explanations: one machine sample → local reason graph.                   |
| SHAP / TreeSHAP                          | Lundberg et al. (2020)<br>Tree explanations from local to global understanding               | Used for tree-based supervised models to show global feature weight and local prediction pushes. |
| PDP                                      | Project topic list allows PDP<br>Visual Analytics requirement: interpretable model behaviour | Shows average marginal effect of one feature on predicted failure probability.                   |

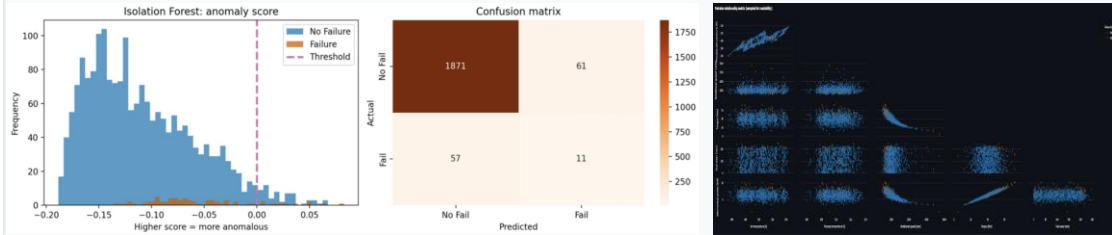
**Defense sentence: supervised XAI explains failure probability; anomaly explanations explain why a sample is abnormal.**

# METHODOLOGY — CRISP-DM CYCLE

## Analytical Approach

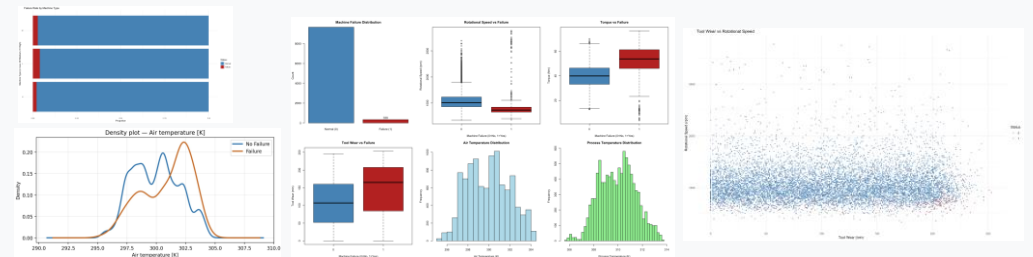
We followed the industry-standard **CRISP-DM Process Model** to build, evaluate, and successfully deploy our predictive maintenance workflow.

Our dashboard incorporates robust exploratory data analysis, allowing engineers to visualize data before triggering machine learning pipelines.



## Dashboard Visualizations Implemented

- ✓ Histograms & Box plots
- ✓ Correlation Heatmaps
- ✓ Density Plots
- ✓ Parallel Coordinates
- ✓ Feature Distributions
- ✓ Real-time telemetry



1

### Business Understanding

Predict machine failures to minimize downtime.

2

**Data Understanding**  
10,000 rows, understand distribution & features.

3

**Data Preparation**  
Address imbalance, scale and split dataset.

4

**Modeling**  
Train Supervised, Anomaly & Semi-Supervised models.

5

**Evaluation**  
Evaluate via ROC-AUC, F1 & Recall.

6

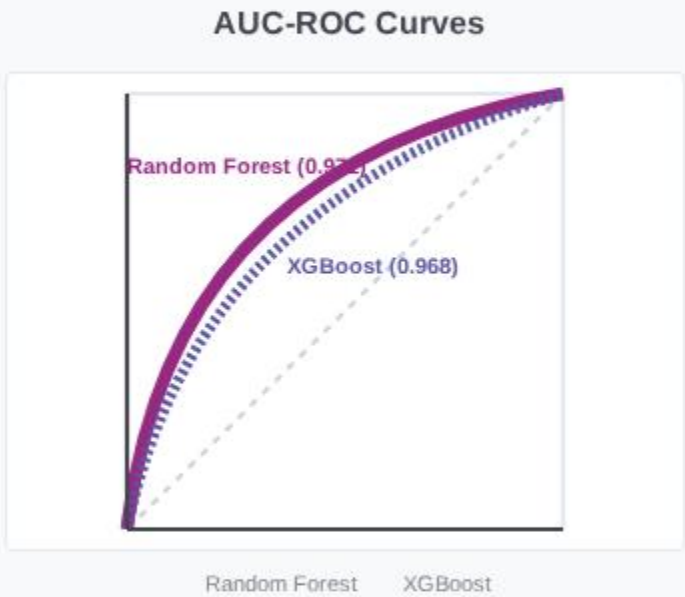
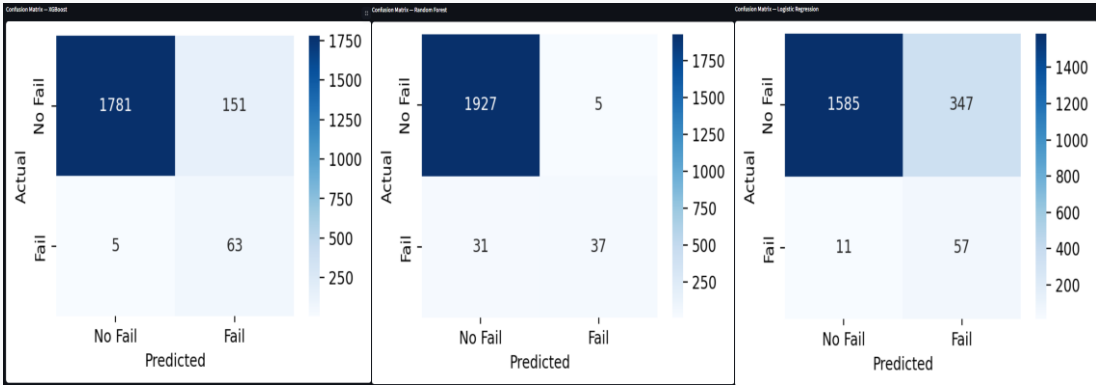
**Deployment**  
Integrate model results in Streamlit Dashboard.

# SUPERVISED MODEL COMPARISON

| Model               | AUC-ROC | Precision | Recall | F1-Score |
|---------------------|---------|-----------|--------|----------|
| Random Forest       | 0.9708  | 0.8810    | 0.5441 | 0.6727   |
| XGBoost             | 0.9682  | 0.2944    | 0.9265 | 0.4468   |
| Logistic Regression | 0.9062  | 0.1411    | 0.8382 | 0.2415   |

## Key Decision Insights

**Random Forest** provides the absolute best precision (0.881) — perfect to minimize false maintenance alarms.  
**XGBoost** excels at recall (0.927) — crucial when the cost of missing any failure is exceptionally high.

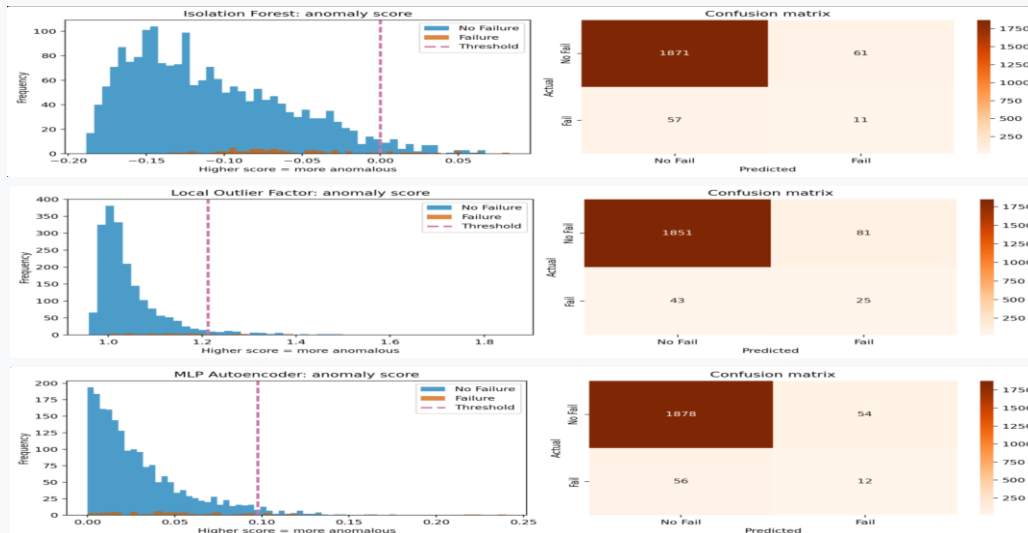


# ANOMALY DETECTION & SEMI-SUPERVISED ML

## Anomaly Detection

Models trained strictly on 100% normal operational behavior:

| Method              | AUC    | P     | R     | F1    |
|---------------------|--------|-------|-------|-------|
| Local Outlier (LOF) | 0.8669 | 0.237 | 0.368 | 0.287 |
| Isolation Forest    | 0.7845 | 0.153 | 0.162 | 0.157 |
| MLP Autoencoder     | 0.7171 | 0.182 | 0.177 | 0.179 |

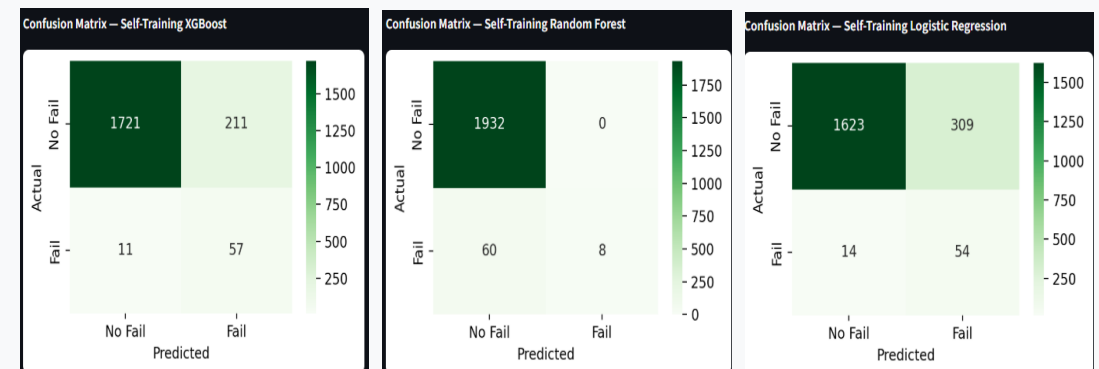


## Semi-Supervised (Self-Training)

Ideal for real industrial use cases where labeling rare failure states is too expensive or logs are completely unlabeled.

**Key Takeaway:** training on only 30% of labeled target data.

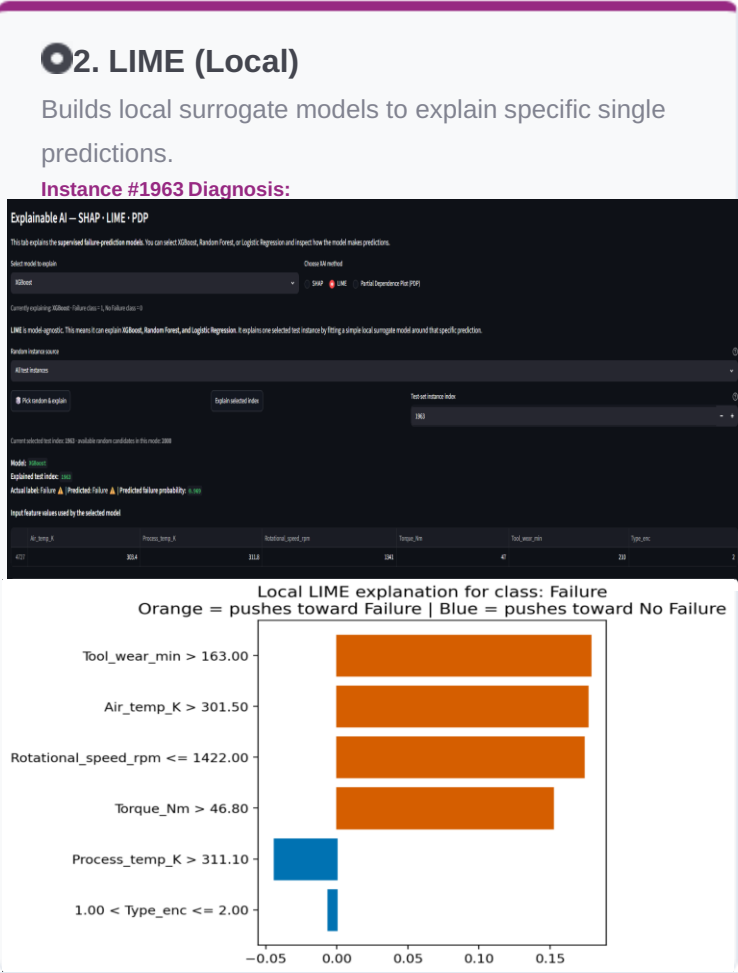
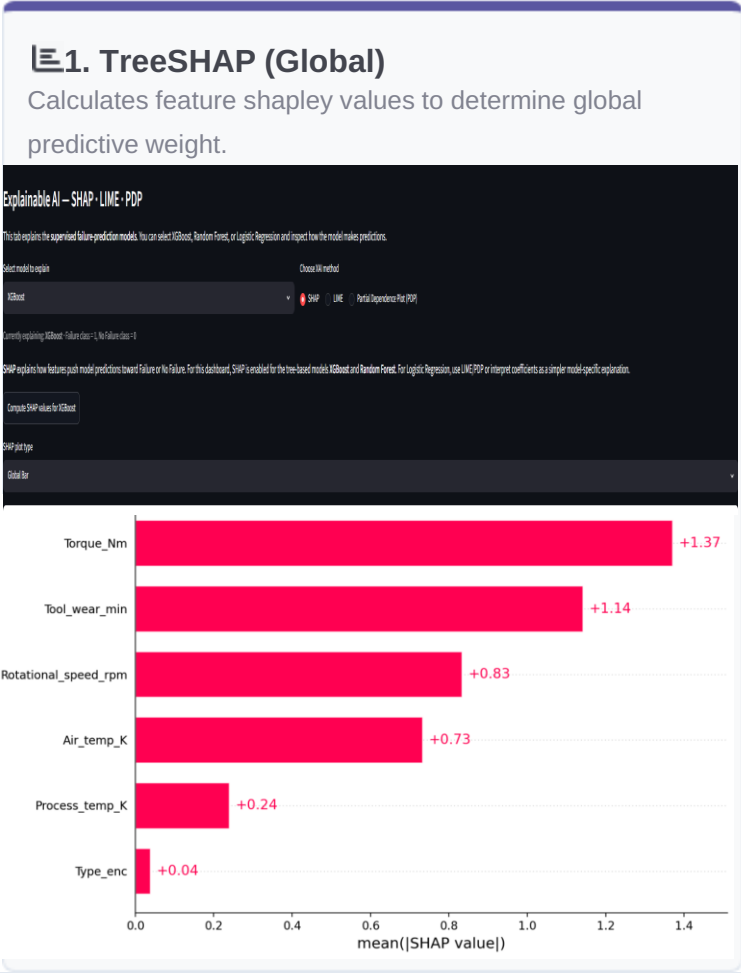
| Method                            | AUC    | P     | R     | F1    |
|-----------------------------------|--------|-------|-------|-------|
| Self-Training XGBoost             | 0.9336 | 0.212 | 0.838 | 0.339 |
| Self-Training Random Forest       | 0.9444 | 1.00  | 0.117 | 0.210 |
| Self-Training Logistic Regression | 0.9009 | 0.838 | 0.794 | 0.250 |



# EXPLAINABLE AI (XAI) FRAMEWORK

Providing predictions is not enough. Machine operators must understand the "why" to trust automated maintenance recommendations, aligning perfectly with the EU AI Act's "Right to Explanation".

SHAP LIME PDP



# INTERACTIVE DASHBOARD DESIGN

## Shneiderman's Mantra UI Design

### 1. Overview

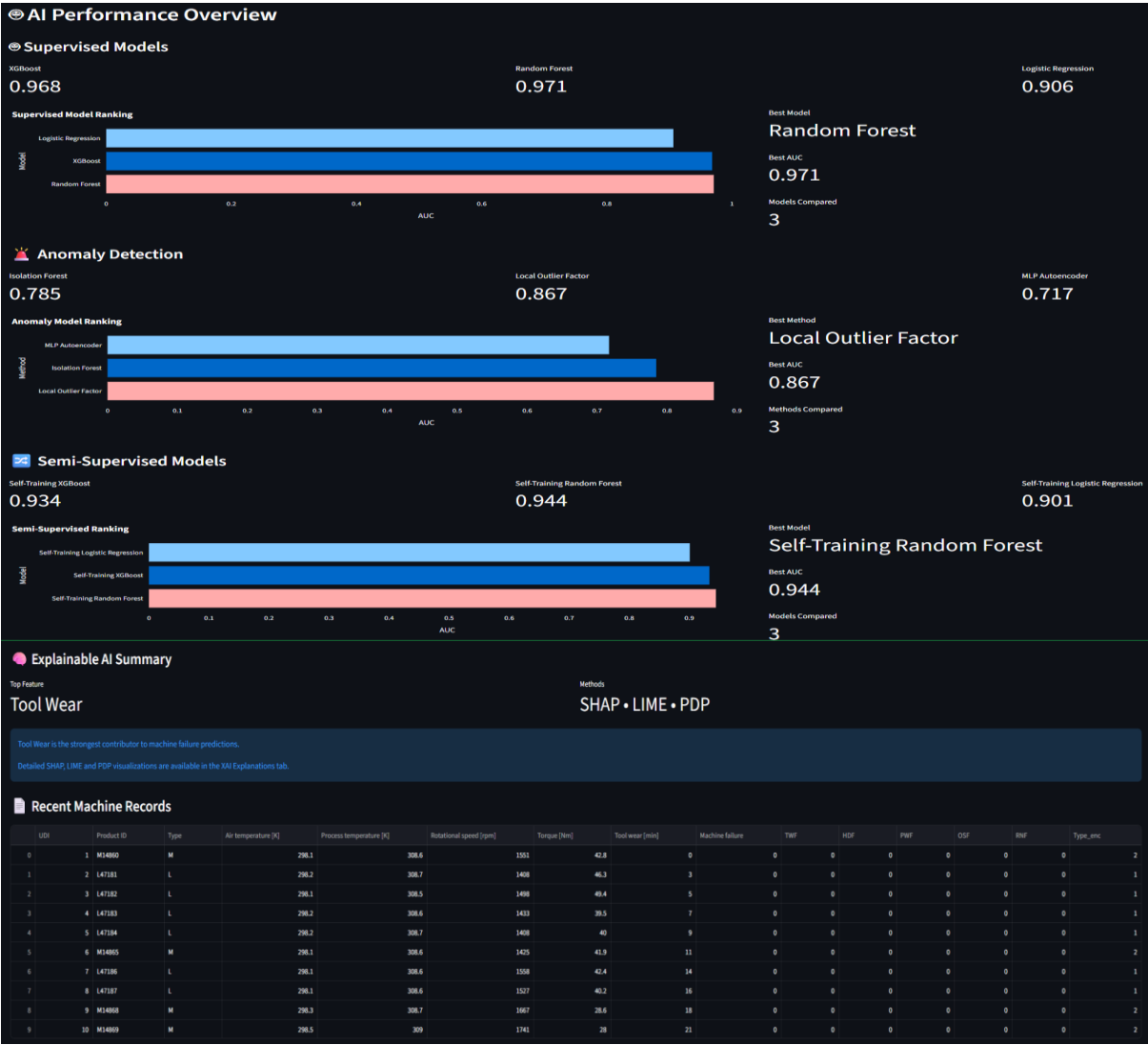
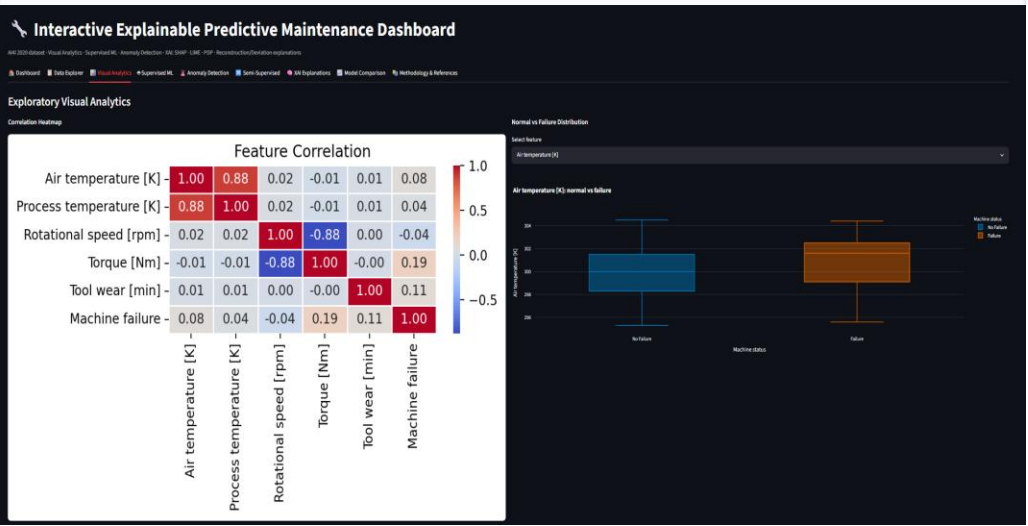
Explore dataset, correlation heatmap, and overall trend charts.

### 2. Zoom & Filter

Filter by machine type (L, M, H) or tool wear ranges.

### 3. Details-on-Demand

Deep dive: Select specific telemetry rows to trigger SHAP / LIME.



# METHODOLOGICAL DECISIONS & REFERENCES

## Robust ML Guidelines

- ✖ **Input Hygiene:** Failure-mode tags (TWF, HDF, PWF) were completely withheld from inputs to prevent analytical data leakage.
- ✔ **Imbalance Handling:** Structured model class weights were utilized instead of artificial SMOTE synthetic sampling.
- ✔ **Verification:** Exact, strictly controlled train/test random splits were shared identically across all compared algorithms.

## Key References

- [1] UCI Machine Learning Repository, **AI4I 2020 Predictive Maintenance Dataset**. DOI: 10.24432/C5HS5C.
- [2] S. Matzka, **Explainable Artificial Intelligence for Predictive Maintenance Applications**, International Conference on Artificial Intelligence for Industries (AI4I), 2020. DOI: 10.1109/AI4I49448.2020.00023.
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- [5] S. M. Lundberg et al., **From Local Explanations to Global Understanding with Explainable AI for Trees**, Nature Machine Intelligence, 2020.
- [6] T. Chen and C. Guestrin, **XGBoost: A Scalable Tree Boosting System**, KDD, 2016.
- [7] L. Cummins et al., **Explainable Predictive Maintenance: A Survey of Current Methods, Challenges and Opportunities**, arXiv:2401.07871, 2024.
- [8] A. Maged, S. Haridy, and H. Shen, **Explainable Artificial Intelligence Techniques for Accurate Fault Detection and Diagnosis: A Review**, arXiv:2404.11597, 2024.
- [9] L. C. Brito, G. A. Susto, J. N. Brito, and M. A. V. Duarte, **An Explainable Artificial Intelligence Approach for Unsupervised Fault Detection and Diagnosis in Rotating Machinery**, arXiv:2102.11848, 2021.

# Thank You!

## Questions & Answers

### ✓ Accomplishments

Built multi-paradigm pipelines (Supervised, Anomaly, Semi-Supervised) paired with XAI (SHAP, LIME, PDP).

### 🔗 UI Paradigm

Applied Shneiderman's Mantra to build user-friendly exploration paths and instant point-level explanations.

### ▶▶ Future Scope

Streaming sensors, counterfactual machine guidelines, and cloud-native dashboard telemetry.